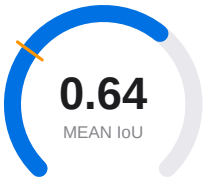
What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

FINDING

419633358 is the best physical fit.

Forward replay tested 6 candidates; the runner-up fits at 0.46. A ranking by physical consistency under tested assumptions — an investigative lead for a human analyst.



MEAN IOU	VARIANTS WON	LEAD RETAINED	CANDIDATES
0.64	**6** / 6	**78%**	**6**
the fit gate is 0.3	forcing variants where it fits best	of 80 tested changes	replayed by forward replay

EVIDENCE

- Best fit under **6 of 6** forcing variants
- Inside the origin area **12 h** before the image
- Ahead of the runner-up by a mean loss gap of **0.1096**
- The nearest vessel, **419427337**, 0.0024° away, is not supported

Candidates, ranked by physical fit

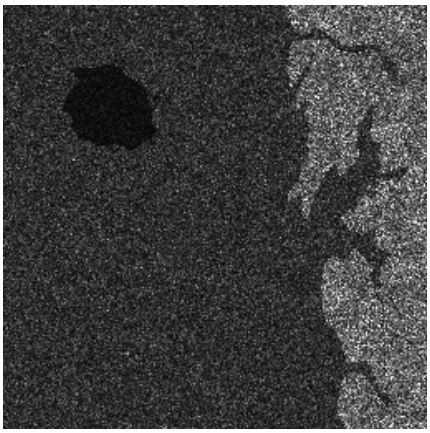
Each candidate's release replayed forward under six forcing variants and compared with the observed slick.

#	VESSEL	MEAN IoU		VARIANTS WON	RELEASE STARTS
1	419633358		0.643	6 of 6	9 Sep 2026, 23:30 UTC
2	419541913		0.464	0 of 6	9 Sep 2026, 17:30 UTC
3	419427337		0.187	0 of 6	10 Sep 2026, 04:00 UTC
4	419924565		0.041	0 of 6	9 Sep 2026, 14:30 UTC
5	419849372		0.005	0 of 6	9 Sep 2026, 10:30 UTC
6	419762703		0.000	0 of 6	9 Sep 2026, 18:30 UTC

The proximity baseline — the vessel nearest the slick when the image was taken — would pick **419427337**, ranked 3 by forward replay.

The scene

The radar image the slick was detected in. Radar darkness alone is not proof of oil.



Acquired	10 Sep 2026, 04:30 UTC
Sensor	synthetic SAR-like (VV proxy, single channel)
Pixel spacing	433.59 m
Data status	synthetic
Label source	generator ground truth (synthetic) — not a human or independently confirmed label
Projection	EPSG:32643

Robustness

Does the leader survive small changes to the model's assumptions?

78%
LEAD RETAINED

80 tested changes to the model's assumptions; **18** changed the leader.

The smallest change that flips it: **diffusivity multiplier 0.6** (magnitude 0.5), under which **419541913** leads.

Single-axis and two-axis perturbations only. Three-or-more-axis combinations were not tested, so absence of a flip is evidence of stability across the tested set, not a proof of stability.

Forward exposure

Where the oil is modelled to drift after the image. Reanalysis-style retrospective field — not an operational forecast.

AFTER THE IMAGE	MODELLED SPREAD	SHORELINE	FISHING GROUND	MANGROVE RESERVE
+6 h 10 Sep 2026, 10:30 UTC	491 km^2	0%	44%	0%
+12 h 10 Sep 2026, 16:30 UTC	545 km^2	0%	40%	0%
+24 h 11 Sep 2026, 04:30 UTC	594 km^2	1%	29%	0%

Traffic filtering

Every vessel in the scene, checked against where the oil could have come from over the 24 hours before the image.

6 Candidate track data present and inside the backward envelope at some queried time

Unassessed is reported separately from excluded: missing evidence is not exoneration.

Coverage-aware matching

Matching radar contacts to AIS when some AIS is hidden — tested on synthetic contacts.

RATE	VESSELS HIDDEN	AREA GOES DARK
Hidden AIS recovered	**85%** 34/40	**84%** 37/44
False alerts	**33%** 17/51	**3%** 1/38
Mistaken promotions	**20%** 8/40	**67%** 12/18

Provenance

AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.
Generated	26 Sep 2026, 17:10 UTC
Software	Python 3.12.14 · Darwin 25.2.0 (arm64)

File integrity

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

attribution_result.json	`b0d2368de1991d0880da7e8f58926a3de55115adde6971dfc3c2a3ec8f5a100e`
coverage_result.json	`8968232d83024e546e55d536113d32cc751890a74c4bc750169c67395254f5e0`
forcing.npz	`d83dc9c6dd317a06e5ab6ca142083c5f10ea299dd52712f37beecb8a556092de`
forecast_result.json	`ab6c8add54f6200f80aae5d4b6e69c2ff12590e4ec2301e4d7528096d79ec370`
fragility_result.json	`e1f2f6d7a0492fd4bde2166ce71c342ac10267c46e2494a5c5d97a8c5e2a4d3b`
historical_mapping.json	`a6d165828c2ecd401974eb76202672e41fd154a0369cb0ca2cd54438c43530b1`
manifest.json	`f1adae2daa13666187698d921c89b794fd6e68443ddebce978b7feb17d4e5a21`
mask_groundtruth.png	`828e0066c72872fa55dcd4ed76493adb320e727d384f7739c0fd9f776340d6f6`
naive_baseline_result.json	`7cef3aa7a6dd370d3c2472c1611eec39243defb153cc782c409b0b3c93523a66`
next_observation_result.json	`c3c79e8587f2673409c99f70404ca120b68230f2ecc039d47395e352b0e63551`
scene.png	`70ea60ae4ce27acc808ea4918e8f08ee08705323f812e85ff2906a236286c179`

Uncertainty

- Rankings express physical consistency under tested assumptions. They are not proof of responsibility.
- Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.
- Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.
- Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

Restated. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.